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SCANMS

SALES COLLABORATOR & AFFILIATE NETWORK MANAGEMENT SYSTEM

CAPSTONE PROJECT DEFENSE | REVIEW 2
Team: FA26SE032
Major: Software Engineering | FPT University

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CONTENT

01 SYSTEM ARCHITECTURE

02 CONCEPTUAL DATA MODEL

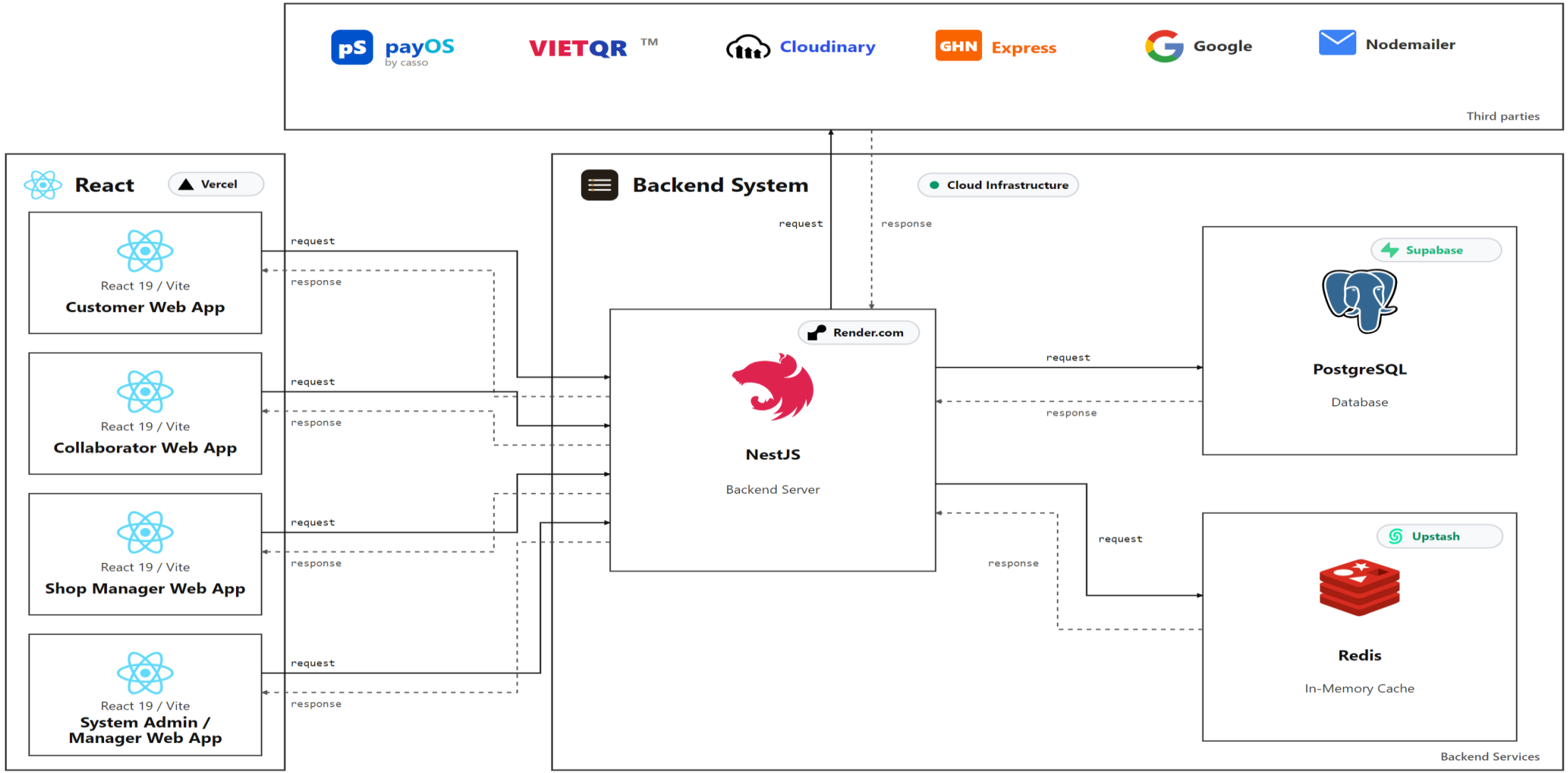
03 FIVE BUSINESS FLOWS & UAT

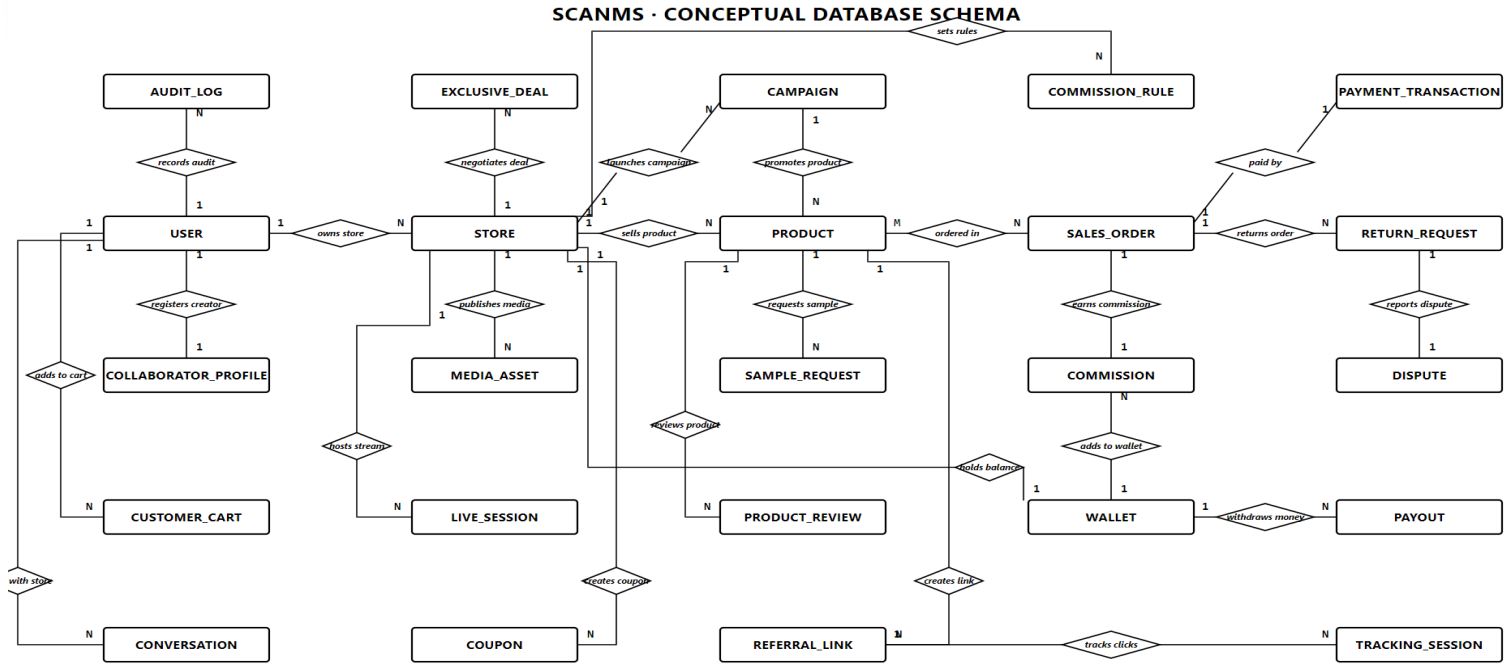
04 Q&A

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SYSTEM ARCHITECTURE




SCANMS · CONCEPTUAL DATABASE SCHEMA
sets rules
AUDIT_LOG
EXCLUSIVE_DEAL
CAMPAIGN
COMMISSION_RULE
PAYMENT_TRANSACTION
records audit
negotiates deal
launches campaign
promotes product
paid by
USER
owns store
STORE
sells product
PRODUCT
ordered in
SALES_ORDER
returns order
RETURN_REQUEST
registers creator
publishes media
requests sample
earns commission
reports dispute
adds to cart
COLLABORATOR_PROFILE
MEDIA_ASSET
SAMPLE_REQUEST
COMMISSION
DISPUTE
reviews product
hosts stream
adds to wallet
holds balance
CUSTOMER_CART
LIVE_SESSION
PRODUCT_REVIEW
WALLET
withdraws money
PAYOUT
with store
creates coupon
creates link
CONVERSATION
COUPON
REFERRAL_LINK
tracks clicks
TRACKING_SESSION

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ACCESS CONTROL

FIVE ROLES, CLEAR RESPONSIBILITIES



CUSTOMER

Marketplace and live shopping
Affiliate purchases
GHN tracking and returns



COLLABORATOR

Identity verification
Smart Links, QR and coupons
14-day samples and commissions



SHOP MANAGER

Six-section products
Samples and VIP deals
Monthly bonuses
Shop wallet reconciliation



SYSTEM MANAGER

KYC and product review
Sample dispute arbitration
Independent return decisions



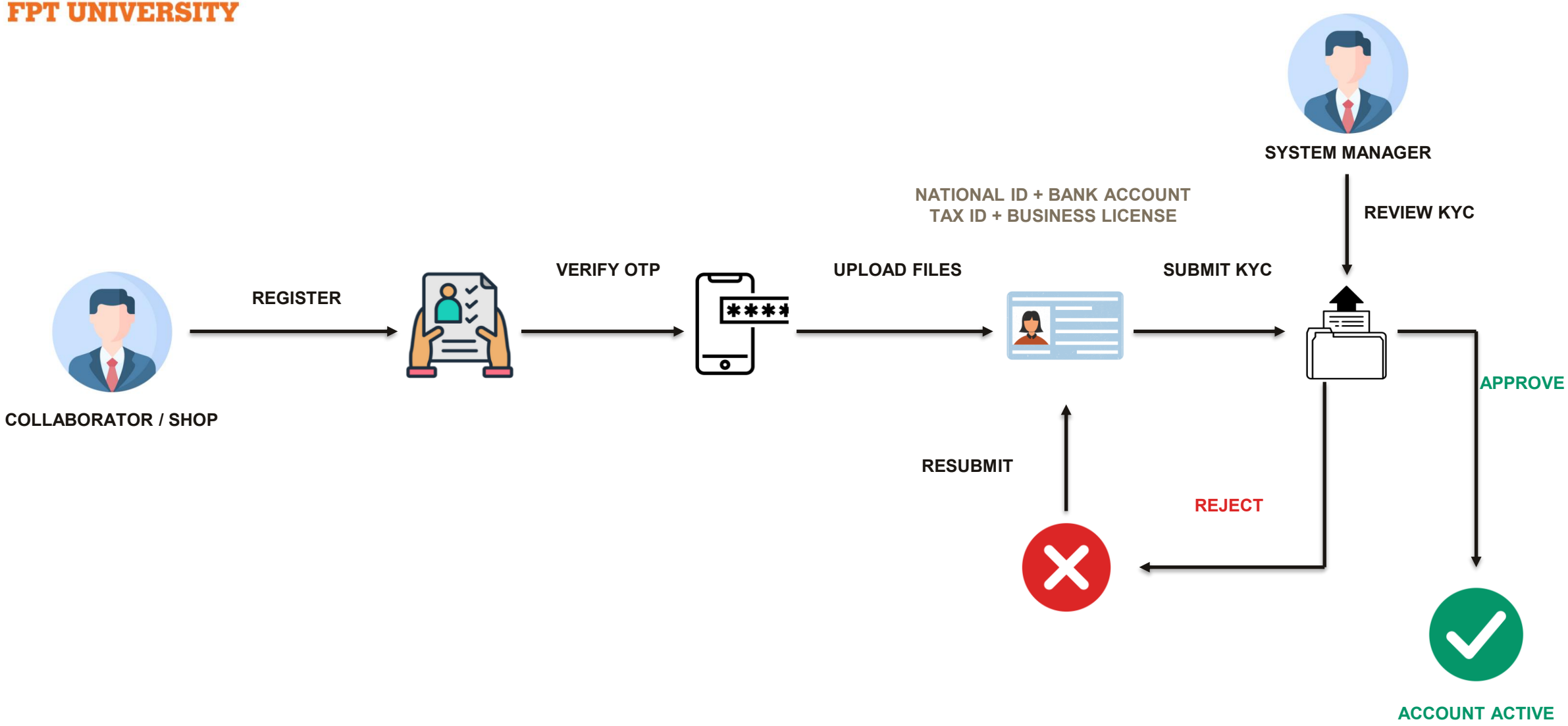
SYSTEM ADMIN

All operations permissions
Exclusive user and role control
Executive financial reports

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FLOW 1: REGISTRATION & KYC



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UAT VALIDATION

FLOW 1: REGISTRATION & KYC

SUCCESS CASES

TC_KYC_01 Account Registration
Enter Name, Email, Phone, Password -> OTP sent.
Registration verified; redirect to secure login.

TC_KYC_02 2FA Login & Role Portal
Submit Email & Password -> Enter 6-digit OTP code.
Secure JWT issued; redirect to role-based portal.

TC_KYC_03 KYC Dossier Submission
Enter 12-digit ID, Bank account, and upload 2 photos.
Dossier submitted; badge marked 'Pending Review'.

TC_KYC_04 System Manager KYC Approval
Manager verifies identity credentials -> Click Approve.
Status set to 'Verified'; full Affiliate features active.

EXCEPTION HANDLING

TC_KYC_01 Duplicate Email / Weak Password
Duplicate Email: Display 'Email already registered'.
Password <8 chars: Block submit, show inline error.

TC_KYC_02 Invalid Password / OTP Limit
Wrong password: Show secure error, protect identity.
>5 wrong OTP: Lock account 15 mins against brute-force.

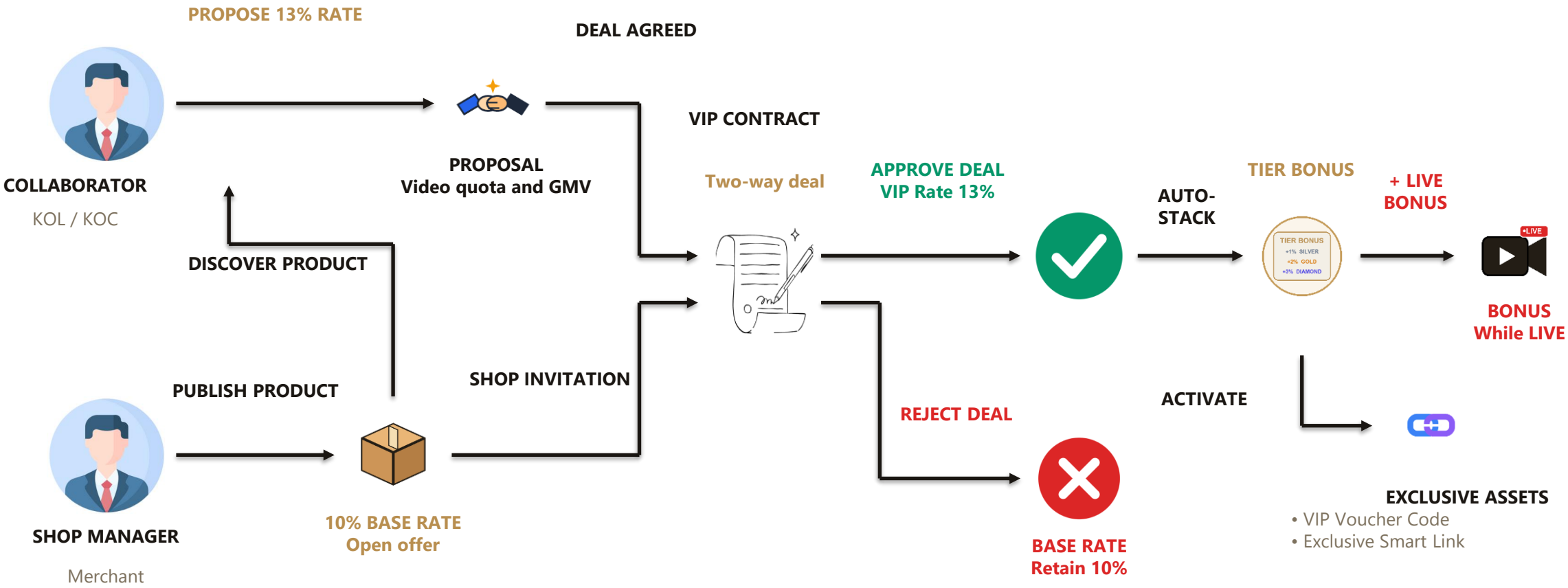
TC_KYC_03 Oversized File / Invalid ID
File >5MB: Reject upload with clear size notification.
ID not 12 digits: Highlight red, disable submit button.

TC_KYC_04 Dossier Rejection with Feedback
Rejecting requires detailed mandatory feedback note.
Notify user to re-upload clear photos; retain account.

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FLOW 2: DEALS & LIVESTREAM



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UAT VALIDATION

FLOW 2: VIP DEALS & LIVESTREAM

SUCCESS CASES

TC_DEAL_01 Exclusive Deal Proposal

Shop selects product, proposes 13% rate & GMV quota.
Deal card instantly appears in KOL chat via Socket.IO.

TC_DEAL_02 Deal Acceptance & VIP Voucher

KOL clicks 'Accept Deal' directly inside chat room.
Deal marked Approved; VIP Voucher & Smart Link issued.

TC_DEAL_03 Live Broadcast & +3% Bonus

Shop pins Deal; KOL starts live video broadcast.
Red Live badge active; commission boosted to 16%.

TC_DEAL_04 Live Session Summary

End stream -> Summary analytics dashboard displays.
Chat & pinned deals locked; rate returns to base 13%.

EXCEPTION HANDLING

TC_DEAL_01 Rate Exceeds Cap (>25%)

Input rate >25%: API blocks submission with warning.
Network disconnect: Display auto-reconnect banner.

TC_DEAL_02 Proposal Rejected / 48h Timeout

KOL clicks Reject: Deal cancelled, retain base 10%.
No response after 48h: Proposal expires, locked.

TC_DEAL_03 Out of Stock / 25% Ceiling Cap

Stock = 0: Hide purchase button in live tray automatically.
Rate calculation: Automatically enforce 25% hard ceiling.

TC_DEAL_04 Tab Closed / Post-Stream Chat

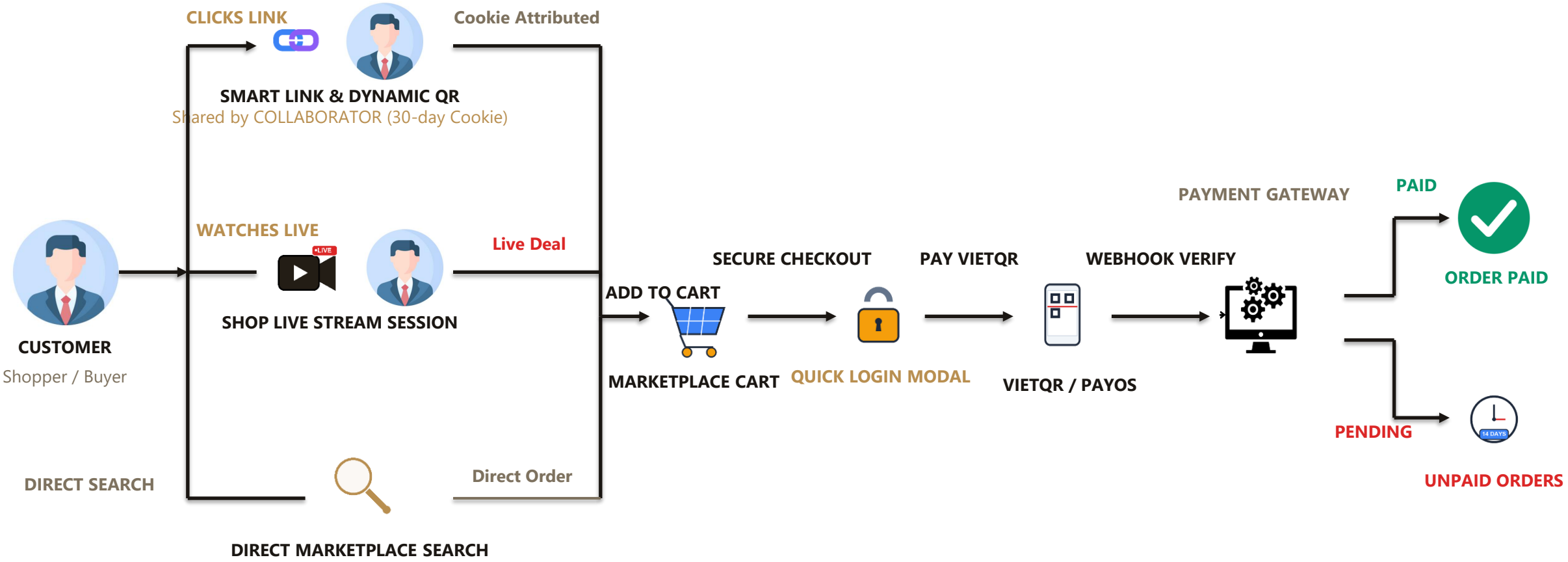
Tab closed: Server safely closes session after 5 mins.
Message after live: Display 'Live session has ended'.

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BUSINESS WORKFLOW

FLOW 3: MULTI-CHANNEL ATTRIBUTION & CHECKOUT



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UAT VALIDATION

FLOW 3: ATTRIBUTION & CHECKOUT

SUCCESS CASES

TC_ATTR_01 Smart Link & Dynamic QR

KOL selects product and clicks 'Generate Link'.
Generates /r/:slug & QR code; download PNG supported.

TC_ATTR_02 30-Day Marketing Cookie

Buyer clicks link -> Redirect to product page.
Cookie stored 30 days; Last-Click updates new KOL.

TC_ATTR_03 In-Place Modal Quick Login

Guest clicks Checkout -> Login modal opens in-place.
Cart 100% preserved after login for instant checkout.

TC_ATTR_04 VietQR Instant Order Matching

Buyer scans VietQR -> PayOS sends instant webhook.
Order marked PAID; commission sent to Escrow Vault.

EXCEPTION HANDLING

TC_ATTR_01 Unverified KYC / Deleted Product

Unverified KYC: Disable link generation with prompt.
Deleted product: Display 404 Discontinued Item page.

TC_ATTR_02 Invalid Slug / Cookie Expired

Invalid slug: Safely redirect buyer to Marketplace.
>30 days: Attributed as organic order without commission.

TC_ATTR_03 Invalid Password / Unregistered

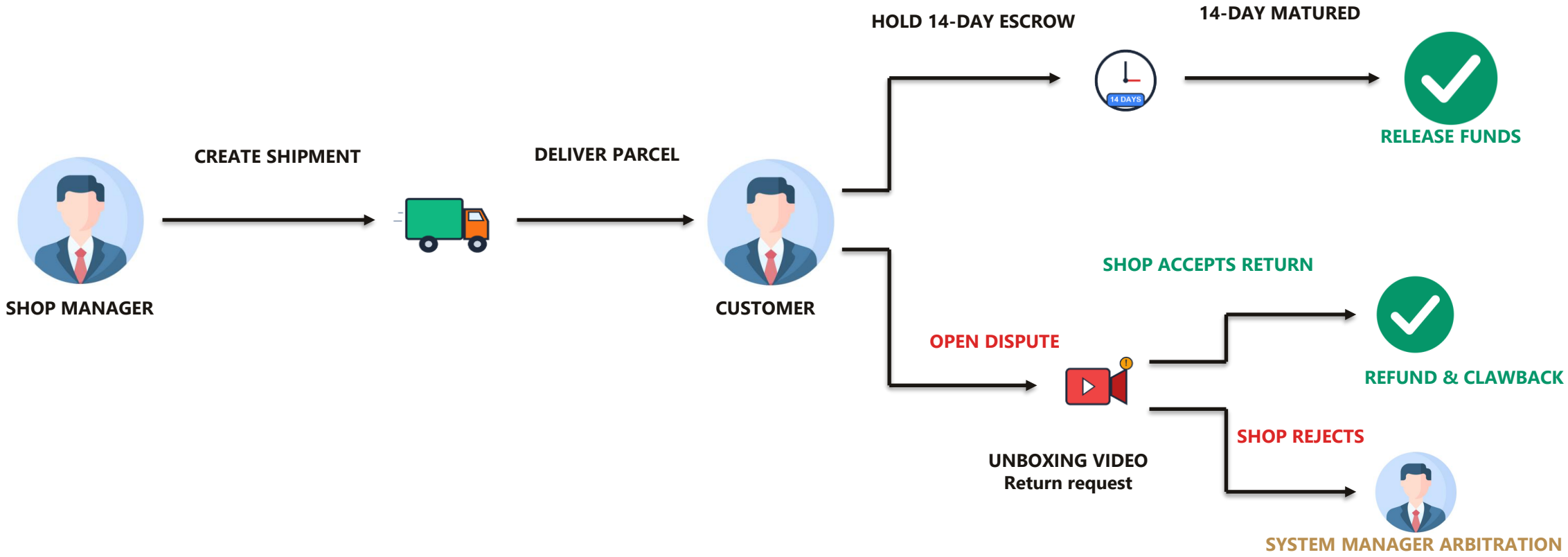
Wrong password: Show inline error without reload.
No account: 30-second Quick Register on same modal.

TC_ATTR_04 Underpaid Amount / Bad Signature

Underpaid: Keep order in Pending Payment tab.
Bad signature: Webhook returns 403, logged to security.

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FLOW 4: FULFILLMENT & DISPUTES



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FLOW 4: FULFILLMENT & DISPUTES

SUCCESS CASES

TC_FUL_01 GHN Waybill Generation
Shop clicks 'Create Waybill' -> GHN API connected.
Tracking code issued; status changed to In Transit.

TC_FUL_02 Delivery & 14-Day Escrow
Delivery successful -> 14-day Escrow clock starts.
Funds secured in Platform Vault; commission protected.

TC_FUL_03 Return Claim with Video
Buyer uploads unboxing video and submits claim.
Order set to Return Requested; Escrow frozen instantly.

TC_FUL_04 Dual-Video Platform Arbitration
System Manager reviews unboxing & packaging videos.
Buyer valid: 100% wallet refund, commission clawed back.

EXCEPTION HANDLING

TC_FUL_01 Missing Dims / Carrier Busy
Missing weight/dims: Block form, prompt input.
GHN maintenance: Display 'Gateway busy, retry later'.

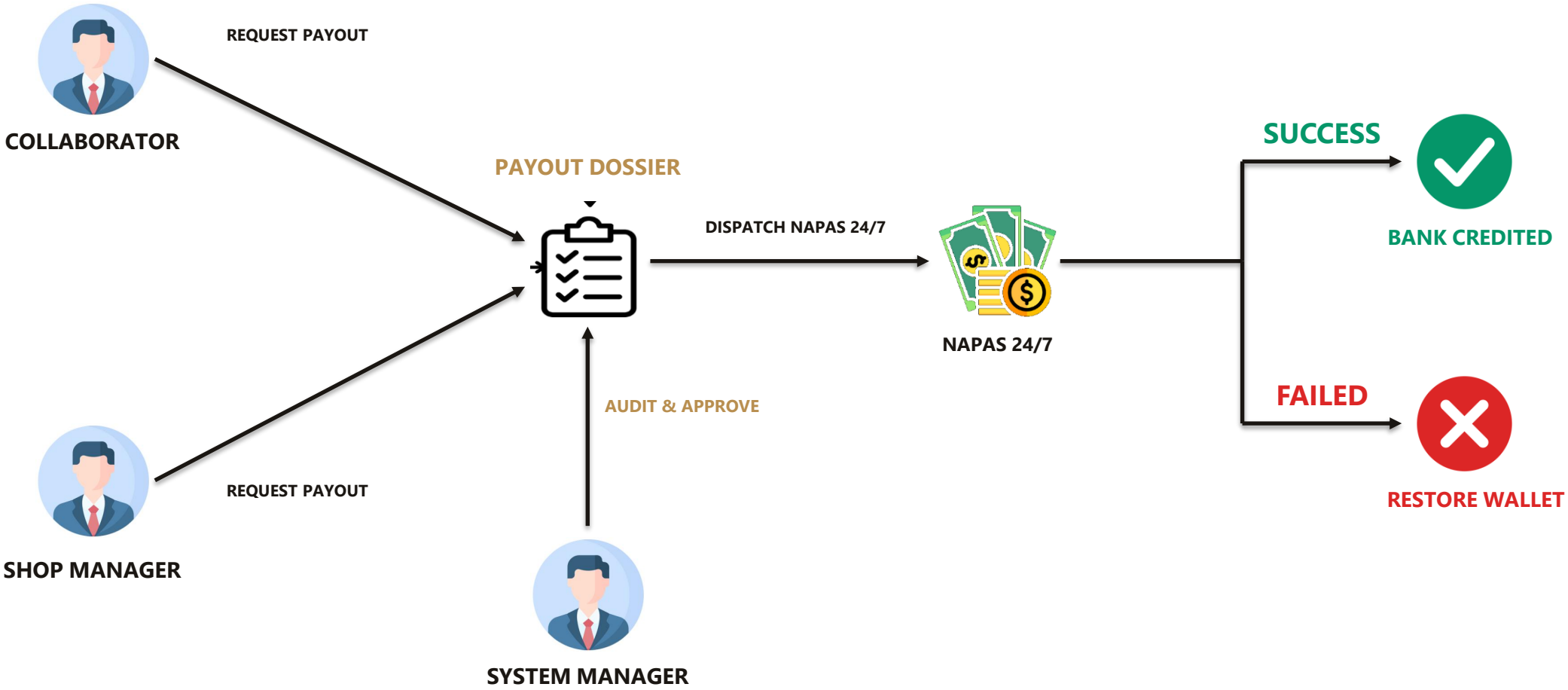
TC_FUL_02 3 Failed Deliveries / Early Payout
3 failed attempts: Return to shop, cancel bonuses.
Withdrawal before 14 days: Alert 'Escrow lock active'.

TC_FUL_03 Expired 14 Days / Missing Video
>14 days delivered: Return button automatically hidden.
Missing unboxing video: Block return claim submission.

TC_FUL_04 Dispute Rejected / Shop Inactive 48h
Buyer invalid: Dismiss dispute, release Escrow to Shop.
Shop inactive 48h: Platform auto-refunds buyer.

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FLOW 5: COMMISSION & REVENUE SETTLEMENT



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UAT VALIDATION

FLOW 5: BONUSES & PAYOUTS

SUCCESS CASES

TC_PAY_01 Monthly Milestone Bonus

Scans monthly net escrow revenue on month-end.
GMV >50M VND earns +2% bonus (e.g. 52.4M * 2%).

TC_PAY_02 Withdrawal Request & Tax Withheld

KOL requests 5,000,000 VND -> 10% PIT tax withheld.
Net 4.5M disbursed; wallet funds locked to prevent double.

TC_PAY_03 Admin Audit & Napas 24/7 Dispatch

System Admin verifies bank details and attaches receipt.
Order marked Disbursed; locked funds permanently deducted.

TC_PAY_04 Automatic Refund on Bank Failure

Bank transfer fails or Admin rejects invalid account.
100% of requested funds safely restored to wallet.

EXCEPTION HANDLING

TC_PAY_01 Returned Orders / Milestone Unmet

Returned orders: Automatically deducted from bonus GMV.
Target unmet: 0 VND bonus, original commission intact.

TC_PAY_02 Insufficient Balance / Below Minimum

Amount > Available balance: Form disables submit.
Amount < 100,000 VND: Alert 'Minimum payout is 100k'.

TC_PAY_03 Unauthorized Access / Missing Receipt

Non-admin user: RBAC immediately blocks with 403.
No bank receipt uploaded: Disburse button disabled.

TC_PAY_04 Bank Gateway Timeout / Stuck Order

Banking timeout: Flag order for reconciliation review.
Never suppress errors; keep funds locked until clear.

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THANK YOU

Q & A

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